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M.Sc. Sem-2
Data Science and Artificial Intelligence (DSAI)/Cyber Security (C.S.)

Experiment No: 5 **Date: / /**

AIM : Interfacing potentiometer with ESP32(Analog Read)

OBJECTIVE: The main objective of this experiment is,

A. To read potentiometer data and print on the Serial Monitor

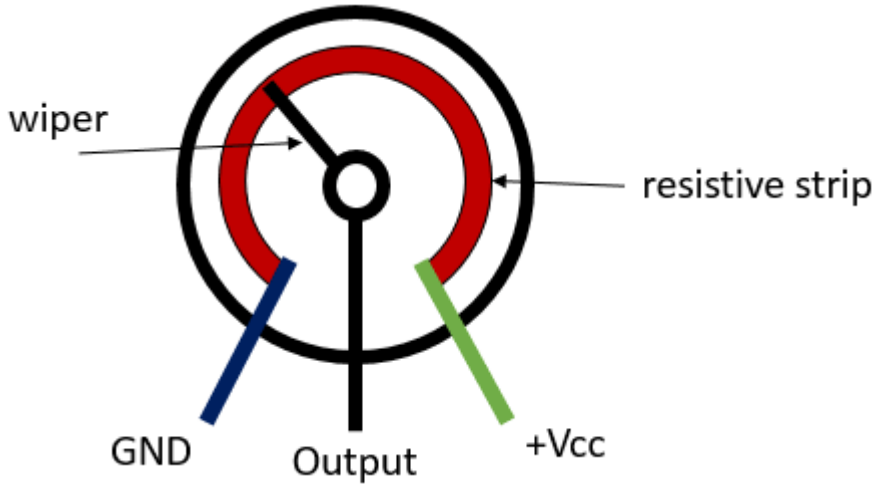
B. To interface potentiometer with LED

APPRATUS: Arduino UNO Board/ ESP32 , LED , Potentiometer, Jumper wires ,breadboard

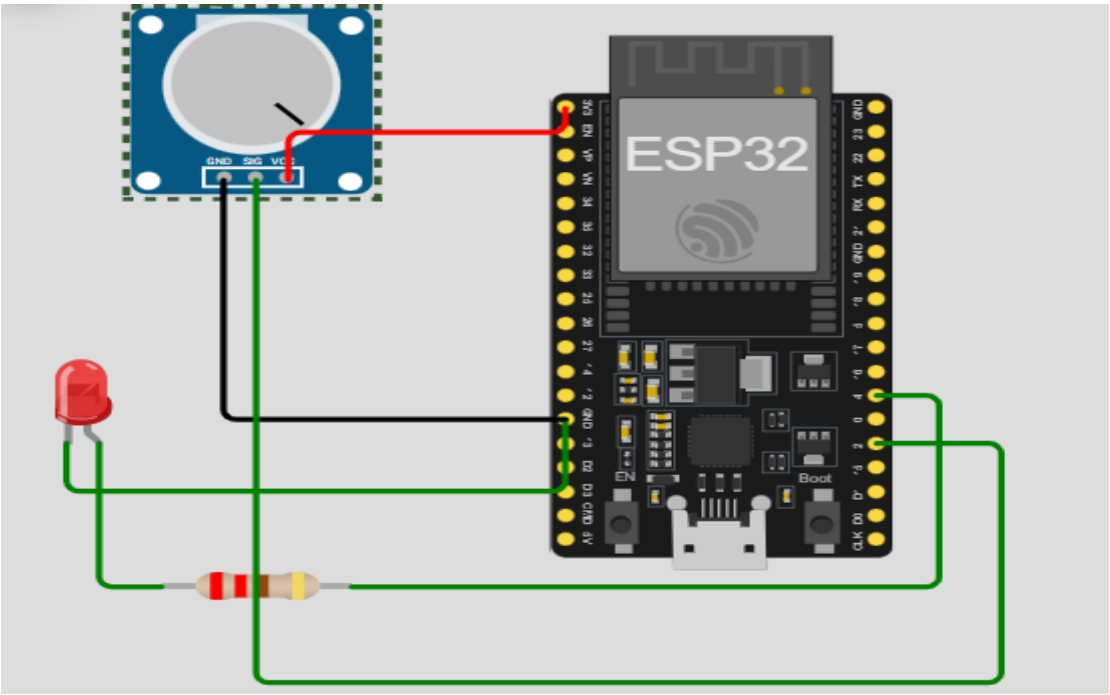
DESCRIPTION:

The potentiometer is a device that is used to measure the voltage or electric potential. It provides a variable resistance when the shaft of the device is turned. The potentiometer is a three-terminal device. It has a rotating contact that acts as an adjustable voltage divider. The potentiometer structure consists of a sliding contact (called wiper), a resistive element, electrical terminals, and housing.

The sliding contact moves along the resistive element, while the housing consists of the wiper and the element.



INTERFACING DIAGRAM:



PROCEDURE:

- Connect the potentiometer to ESP32 Board.
- Connect one terminal of potentiometer to GND
- Connect the other terminal to 5V supply.
- Connect the middle terminal to the ANALOG pin()
- Connect the LED at pin 2
- Change the value of the potentiometer and observe the LED brightness

PROGRAM:

RESULTS:

CONCLUSION:

SAMPLE VIVA-VOCE QUESTIONS:

1) What is the purpose of potentiometer ?
2) What are the parts of potentiometer in ESP32?
3) What are the two main types of potentiometers?
4) How to control potentiometer with ESP32?